

Product Requirements Document (PRD)

Scenario 3: AI Cultural Tourism Insights & Engagement Platform

Note : Use this PRD template to document your research, planning, and progress throughout your Capstone Project. Complete all sections carefully and submit the finalized version to the evaluation panel.

Duration: 10 Weeks

1. Purpose

Define the main purpose of your AI Cultural Tourism Insights & Engagement Platform. Describe how your system applies AI, geospatial data, and language technologies to design customized cultural travel experiences. Explain how it brings innovation by automatically creating multilingual itineraries, smart travel suggestions, and personalized trip videos that adjust to traveler interests, seasons, and feedback.

Discuss how your platform tackles real challenges in the travel and tourism sector for example, how AI-based personalization can make travel planning easier, promote cultural awareness, support sustainable tourism, and help tourism organizations better understand traveler behavior. Mention any surveys, articles, or travel reports you reviewed while researching the problem and explain why your idea is relevant and useful in today's travel-tech industry.

Summarize the strategic intent and tourism-tech relevance of your application below:

This platform uses AI to create personalized cultural travel experiences through smart recommendations, multilingual itineraries, and trip videos. It makes travel planning easier, more engaging, and more relevant to each traveler's interests, season, and preferences.

2. Objectives

Clearly outline what your AI Tourism Platform aims to achieve. Include how you are aiming to achieve below:

- **Personalized Itinerary Generation:** Automatically create travel plans based on traveler inputs such as age, interests, accessibility needs, budget, season, and tour duration.
- **AI-Powered Recommendations:** Use Gemini API and trained models to suggest relevant cultural, adventure, cuisine, nature, or wellness experiences tailored to traveler context.

- **Multilingual Communication:** Generate travel insights, chatbot replies, and captions in multiple languages for global inclusivity.
- **Multimedia Content Creation:** Produce **visually engaging PDFs** and **recap videos** summarizing trip highlights with weather, site data, and travel summaries.
- **Adaptive Feedback Learning:** Collect user feedback on itineraries, chatbot, and recommendations to continuously refine personalization accuracy.
- **Data-Driven Insights:** Visualize tourist patterns and preferences via dashboards powered by Plotly or Tableau.

Specify your measurable goals and project outcomes below:

The platform aims to deliver personalized travel planning through AI-generated itineraries, smart destination recommendations, multilingual support, and engaging PDF/video outputs. It also uses user feedback and tourism data insights to improve accuracy, inclusivity, and overall travel experience quality.

3. Scope

Describe what your project will include, for example:

- **Personalized Itinerary Generation Module** (age, interests, climate, accessibility).
- **Smart Recommendation System** with multilingual descriptions and site images.
- **PDF Export System** for downloadable, multilingual itineraries with visuals.
- **Return Trip Video Generator** using MoviePy/OpenCV.
- **Multilingual Chatbot** for live cultural and travel support.
- **Feedback & Analytics Dashboard** for data visualization and model refinement.

Define your project's boundaries and inclusions/exclusions below:

The project includes modules for personalized itinerary generation, AI-based destination recommendations, multilingual chatbot support, PDF itinerary export, trip recap video generation, and feedback-based analytics. Its scope is limited to cultural tourism planning and user engagement within the platform, and does not include real-time booking, payments, transport reservations, or full-scale commercial deployment.

4. Stakeholders

List the key individuals and groups (end users) who will interact with or benefit from the platform:

For example:

- **End Users:** Tourists, travel planners, and cultural enthusiasts who seek personalized itineraries and interactive AI travel support.
- **Travel Agencies / Tourism Boards:** Potential business adopters to promote cultural destinations effectively.

Describe stakeholder roles and value contributions below:

Key stakeholders include tourists, cultural travelers, and travel planners who use the platform for personalized recommendations, itineraries, multilingual support, and trip outputs. Tourism boards and travel agencies can benefit by understanding traveler preferences better and promoting destinations more effectively through AI-driven insights.

5. Deliverables

Summarize the tangible outputs your project will produce For example:

- **Streamlit Web App:** A deployed, interactive platform integrating all AI modules.
- **Personalized Itinerary Generator:** Curates site sequences with weather and cultural context.
- **AI Recommendation System:** Suggests destinations dynamically with visuals and multilingual text.
- **Multilingual Chatbot:** Provides travel assistance and collects user feedback in real time.
- **PDF Exporter:** Creates visually rich itinerary documents with images, site info, weather icons, and text in multiple languages.
- **Trip Recap Video Generator:** Produces multilingual, music-backed travel videos using MoviePy/OpenCV.
- **Feedback Dashboard:** Displays analytics on traveler ratings, preferences, and satisfaction metrics.

Outline your project deliverables and their expected outcomes below:

The project will produce a Streamlit-based AI tourism platform with modules for personalized itinerary generation, smart destination recommendations, multilingual chatbot assistance, PDF itinerary export, trip recap video generation, and feedback analytics. These deliverables aim to create a complete, interactive travel-planning experience that is personalized, multilingual, and user-centered.

6. Tools and Technologies

Provide a structured overview of your tech stack why and how you have used the tools and respective in the project:

For example:

- **Python & Google Colab:** Core environment for model building, data processing, and experimentation.
- **Streamlit Cloud:** For developing and deploying the interactive AI travel interface.
- **OpenCV & MoviePy:** For video synthesis, image handling, and multimedia storytelling.

Describe preprocessing, feature creation, and dataset integration processes below:

The project uses Python as the core development language, Streamlit for building the interactive web platform, Pandas and NumPy for dataset cleaning and preprocessing, and Gemini API for AI-based recommendations, multilingual responses, and itinerary generation. ReportLab is used for PDF creation, while MoviePy and image-processing tools support trip recap video generation. The dataset was preprocessed by handling missing values, removing duplicates, standardizing fields such as category, season, and location, and organizing features so they could be used effectively for recommendation, itinerary, and analytics modules.

7. Success Metrics

Define measurable indicators of success of our platform:

- **Itinerary Relevance:**How effectively does the platform generate travel plans that align with each traveler's preferences, interests, duration, and seasonal suitability?
- **Recommendation Quality:** Are the suggested destinations, cultural sites, and experiences meaningful, engaging, and personalized to the traveler's profile?
- **Chatbot Effectiveness:** Does the multilingual chatbot provide accurate, quick, and context-aware responses that genuinely assist travelers in planning or adjusting their trips?
- **User Satisfaction:** Are travelers finding the platform easy to use, visually clear, and helpful in enhancing their overall travel experience?

- **System Reliability:** Is the platform performing smoothly with minimal delays, stable multilingual output, and seamless integration between all modules?

List your performance KPIs and measurement approach below:

Success will be measured through itinerary relevance, recommendation quality, chatbot response usefulness, user satisfaction, and overall system reliability. Key KPIs include how well outputs match user preferences, feedback ratings on recommendations and itineraries, chatbot helpfulness, ease of use, and smooth performance across modules with minimal errors or delays.

8. Weekly Roadmap (10 Weeks)

Week 1: Problem Definition & Data Understanding

Define project objectives, analyze datasets (Cultural Tourism, Worldwide Cities, and Destinations), and generate initial data visualizations using Matplotlib.

Planned Tasks:

Define the project problem, finalize objectives, understand the tourism datasets, and perform initial exploratory data analysis with basic visualizations.

Achieved Tasks:

Clearly defined the project scope and objectives, reviewed the Cultural Tourism, Worldwide Cities, and Destinations datasets, and generated initial visual insights to understand patterns, categories, and destination distribution.

Week 2: Data Cleaning & Feature Engineering

Clean and merge datasets; engineer features such as Ideal_Season, Accessibility_Level, and Experience_Type; visualize cultural relevance vs. climate and cost.

Planned Tasks:

Clean and merge the tourism datasets, handle missing values and duplicates, create useful features such as season, accessibility, and experience type, and visualize relationships

between cultural relevance, climate, and cost.

Achieved Tasks:

Cleaned and organized the datasets, removed inconsistencies, standardized important fields, created key travel-related features for personalization, and produced visualizations to better understand destination suitability across climate, culture, and budget factors.

Week 3: Traveler Input & Personalization Engine

Develop a Streamlit input form (age, interests, accessibility, budget, preferred weather). Use Gemini API for multilingual personalization logic and summary text.

Planned Tasks:

Develop the Streamlit traveler input form to collect details like age, interests, accessibility needs, budget, and weather preference, and connect Gemini API for multilingual personalization support.

Achieved Tasks:

Built the personalization input flow in Streamlit, captured key traveler preferences, and integrated Gemini API to support AI-based multilingual summaries and personalized travel assistance.

Week 4: Itinerary Generation Module

Automate itinerary creation (day-wise cultural routes) including site descriptions, weather, and visuals. Display itineraries interactively on Streamlit with translation support.

Planned Tasks:

Develop the itinerary generation module to automatically create day-wise travel plans with cultural site details, weather context, visuals, and translation support in Streamlit.

Achieved Tasks:

Built the itinerary generation flow to produce structured day-wise travel plans, displayed

them interactively in Streamlit, and added AI-supported content and multilingual output support.

Week 5: Smart Recommendation Engine

Implement content and location-based recommender using dataset attributes. Generate AI-driven descriptions and multilingual insights using Gemini API.

Planned Tasks:

Build a smart recommendation engine using destination attributes and location-based matching, and generate AI-powered descriptions and multilingual insights with Gemini API.

Achieved Tasks:

Implemented the recommendation system to suggest relevant destinations based on traveler preferences and dataset features, and integrated Gemini API to generate personalized destination descriptions and multilingual travel insights.

Week 6: PDF Itinerary Export

Use ReportLab/FPDF to generate comprehensive, multilingual PDF itineraries with weather icons, images, and daily highlights. Add download functionality in Streamlit.

Planned Tasks:

Develop a PDF export module to create multilingual itinerary documents with daily highlights, weather details, visuals, and download support in Streamlit.

Achieved Tasks:

Implemented the PDF itinerary export feature using PDF libraries (ReportLab), generated structured downloadable itineraries with key travel details, and integrated the download option into the Streamlit application. Added Itinerary ID to the PDFs so that users can rate it in a new session.

Week 7: Return Trip Video Generator

Use MoviePy/OpenCV to create AI-generated video recaps featuring sites, countries, best seasons, and captions in multiple languages. Integrate optional background music. Add smooth transitions and background music, then render the final video in MP4 format with a Streamlit preview.

Planned Tasks:

Develop a trip recap video generator using MoviePy/OpenCV to create multilingual travel videos with destination visuals, captions, transitions, seasonal highlights, optional background music, and MP4 preview support in Streamlit.

Achieved Tasks:

Built the video generation module to create travel recap videos with destination-based visuals, AI-generated captions, and MP4 output using MoviePy.

Week 8: Multilingual Chatbot Assistance

Integrate Gemini API chatbot for real-time, multilingual interactions. Enable itinerary modifications, destination queries, and feedback submission.

Planned Tasks:

Integrate a multilingual Gemini API chatbot for real-time travel assistance, itinerary updates, destination-related queries, and feedback collection.

Achieved Tasks:

Implemented the AI chatbot module with multilingual support, enabled interactive travel guidance and destination assistance, and connected it with itinerary and feedback-related user interactions.

Week 9: Feedback Analytics & Adaptive Learning

Collect and visualize feedback data (chatbot, itineraries, recommendations) using Plotly/Tableau. Implement adaptive learning to enhance model personalization.

Planned Tasks:

Build a feedback analytics module to collect and visualize user responses on recommendations, itineraries, and chatbot performance, and use this data to improve personalization.

Achieved Tasks:

Implemented feedback collection across major modules, organized user response data for analysis, and added adaptive improvement logic to support better personalization and recommendation quality over time.

Week 10: Integration, Testing & Deployment

Integrate all modules (Itinerary, Recommendation, PDF, Video, Chatbot, Feedback) into one Streamlit app. Conduct system testing, evaluate usability, and deploy the final version on Streamlit Cloud.

Planned Tasks:

Integrate all major modules into one Streamlit application, test the complete system for usability and performance, and prepare the final platform for deployment.

Achieved Tasks:

Combined the itinerary, recommendation, PDF, video, chatbot, and feedback modules into a unified Streamlit app, carried out testing and bug fixing, and finalized the platform in a deployable form.

9. Risk Management

Identify potential technical and operational risks For example does your product address the issue like below and how you imitated it:

- Incomplete or Biased Data
- Slow API Responses
- Multilingual Translation Errors
- Video Generation Failures
- Integration Conflicts

Describe mitigation strategies and monitoring methods below:

Potential risks include incomplete or biased tourism data, slow API responses, multilingual output errors, video generation failures, and integration issues between modules. These were mitigated through data cleaning and preprocessing, fallback logic, error handling, modular testing, and repeated debugging across features. Monitoring was done through system testing, output checks, and user feedback can be used to identify and improve weak points.

10. Evaluation Criteria

Does your project fully assess the following dimensions:

- **Technical Performance:** Model accuracy, integration quality, and dashboard functionality.
- **Innovation:** Creative and practical application of AI, NLP, and multimedia generation.
- **Design & Usability:** Clarity, interactivity, and visual experience of the Streamlit interface.
- **Business & Cultural Impact:** Real-world potential for travel personalization and engagement.
- **Deployment Quality:** A fully functional, stable, and responsive end-to-end system.

Outline the criteria and performance standards used for final assessment below:

The project is evaluated based on technical performance, innovation, design, usability, business and cultural impact, and deployment quality. Assessment focuses on how well the platform integrates AI features, how effectively it personalizes travel experiences, how user-friendly and interactive the interface is, and how stable and complete the final end-to-end system performs in practice.

Final Reflection

Reflect on your learning throughout the Capstone journey. Discuss your experience in developing AI-based personalization, multilingual systems, and multimedia travel solutions. Highlight what you learned about teamwork, innovation, and the future potential of AI in tourism:

This capstone journey helped me understand how AI can be applied in a practical way to solve real-world problems in tourism. I learned a lot about building personalization systems, working with multilingual AI features, and combining recommendation, itinerary, PDF, chatbot, and video modules into one platform. It also improved my teamwork, problem-solving, and debugging skills, especially while managing integration challenges and refining the user experience. Overall, the project showed me the strong future potential of AI in tourism for creating smarter, more personalized, and more engaging travel experiences.